

EOLLM Task Ablation Study

Qwen3.5-4B-VL on EOLLM urban VQA — Apr 26–30, 2026

Ezel Bayraktar

TL;DR

- Removing tasks does not improve kept tasks. Every ablation underperforms PC-base on its own kept subset.
- Withheld-task transfer is partial and asymmetric. mismatch_mcq + camera_direction collapse to chance when their family is fully withheld.
- Withholding only the *hard* half of a family (Split-6) preserves >80% on the hard variant: easy/hard mismatch share representation.
- Withholding camera_direction alone (Split-5) leaves it at chance (29.9%, random=25); zero transfer from urban/mismatch supervision.
- Cross-city baseline (SU-base) is genuinely harder: 70.8% vs PC-base 78.3%.
- green_space and road_surface stay at ~94–100% in every PC run — leakage, not learning.
- Split-2a regressed: ep1 59.5% → ep3 50.7%. Removing both mismatch and camera_dir destabilizes training.
- More epochs do not close the withheld-task gap (Split-1 plateaus by ep4).

1 Run matrix

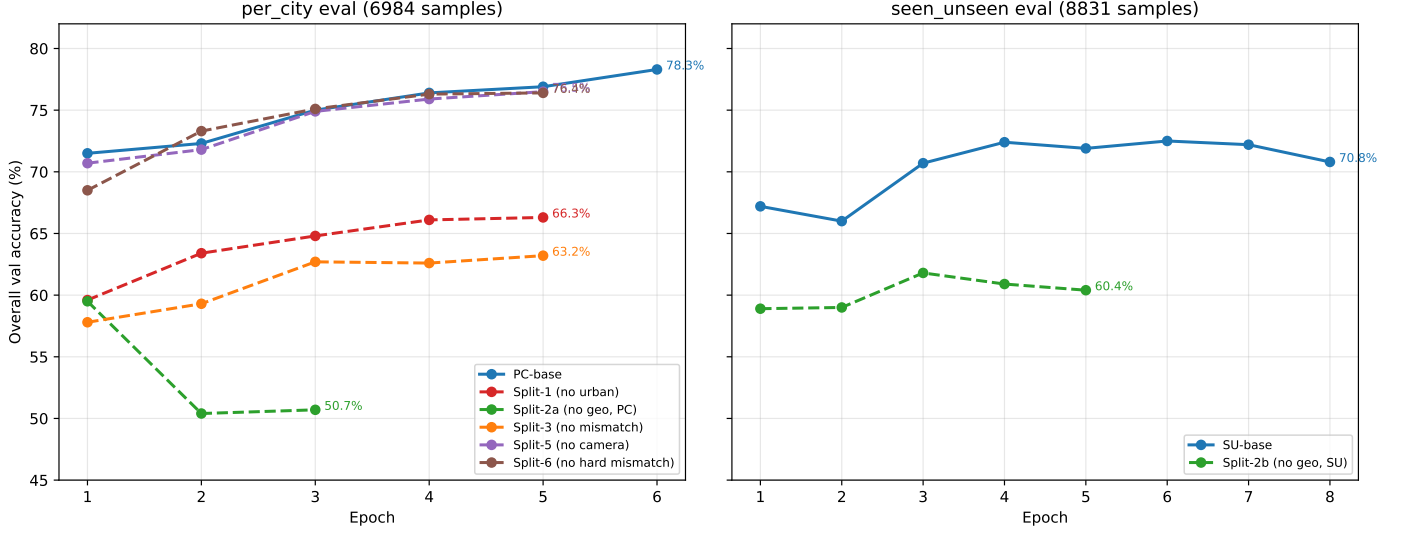
Run	Split	Topics removed from training	Epochs	Status
PC-base	per_city	none	6	done
SU-base	seen_unseen	none	8	done
Split-1	per_city	7 urban: amenity, bldg_h, junction, land_use, road_type, transit, urban_density	5	done
Split-2a	per_city	5 geo: camera_direction, all 4 mismatch_*	3	done (regressed)
Split-2b	seen_unseen	5 geo: camera_direction, all 4 mismatch_*	5	done
Split-3	per_city	4 mismatch_* only (camera_dir kept)	5	done
Split-5	per_city	camera_direction only	5	done
Split-6	per_city	mismatch_binary_hard + mismatch_mcq_hard (easy variants kept)	5	done

Eval = all 14 topics in every run. bs=96, LoRA r=16/ α =16, lr=2.5e-4, RTX Pro 6000 96 GB.

2 Headline

- PC-base ends at 78.3%, still rising.
- Every per_city ablation underperforms the full baseline on the same eval.
- SU-base (70.8%) is the honest cross-city number, ~7.5 pp below PC-base.

Overall validation accuracy by epoch



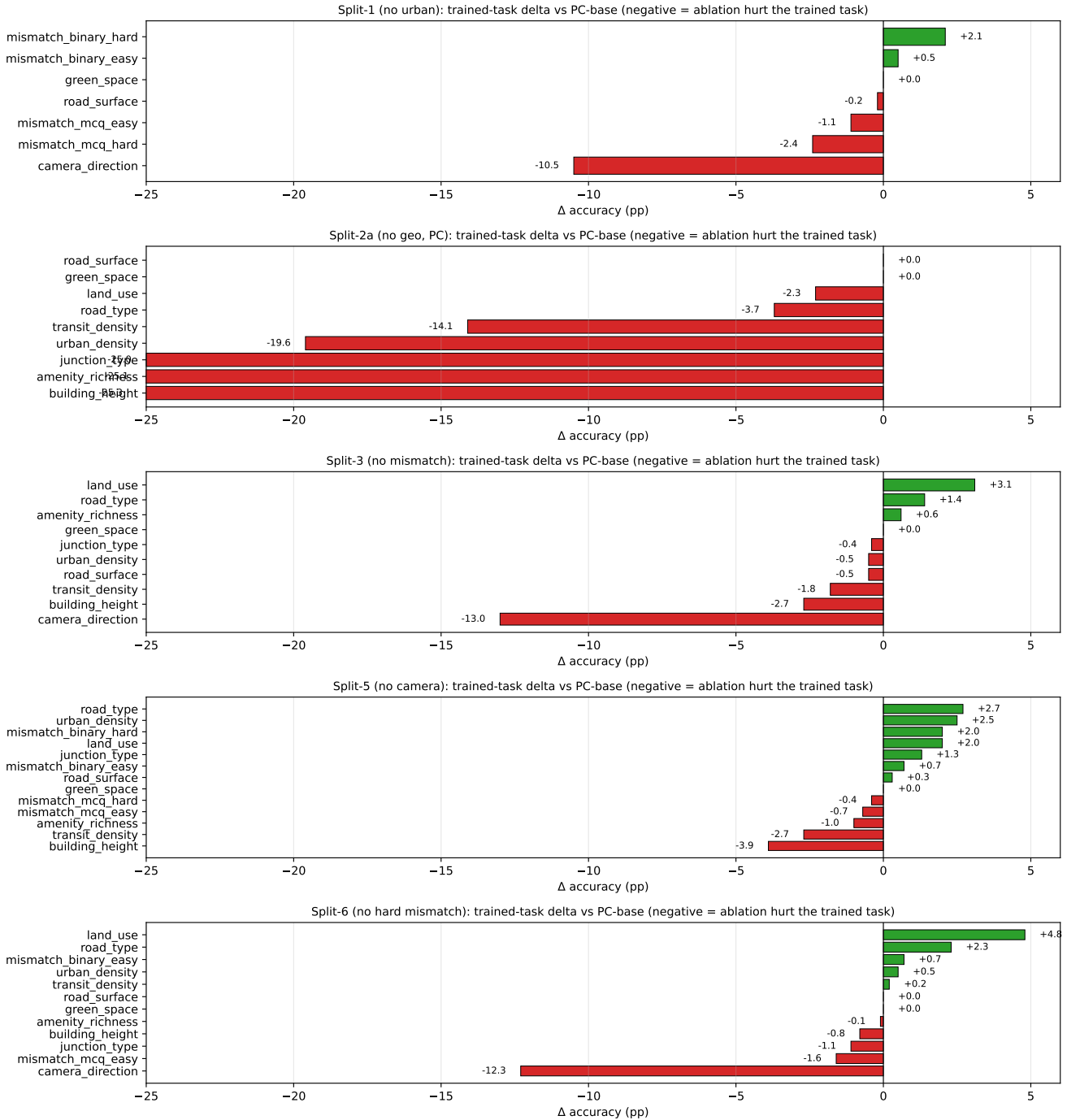
3 Per-topic final accuracy

Topic	PC-base full	SU-base full	Split-1 no urban	Split-2a no geo PC	Split-2b no geo SU	Split-3 no mism.	Split-5 no cam	Split-6 no hard mism.
amenity_richness	62.0	59.4	38.0	36.9	59.7	62.6	61.0	61.9
building_height	69.0	54.4	52.1	43.7	54.0	66.3	65.1	68.2
junction_type	77.0	71.4	60.7	52.0	69.6	76.6	78.3	75.9
land_use	75.9	69.1	50.8	73.6	76.2	79.0	77.9	80.7
road_type	71.3	70.9	55.3	67.6	72.1	72.7	74.0	73.6
transit_density	50.8	36.9	31.0	36.7	44.9	49.0	48.1	51.0
urban_density	73.8	70.6	41.7	54.2	72.9	73.3	76.3	74.3
camera_direction	56.5	55.6	46.0	26.2	26.1	43.5	29.9	44.2
mismatch_binary_easy	96.6	95.0	97.1	54.5	77.4	75.2	97.3	97.3
mismatch_binary_hard	90.2	91.6	92.3	54.7	68.1	65.2	92.2	82.7
mismatch_mcq_easy	99.1	98.5	98.0	26.6	37.3	31.4	98.4	97.5
mismatch_mcq_hard	90.6	90.5	88.2	23.4	31.1	29.4	90.2	81.6
green_space	100.0	63.8	100.0	100.0	100.0	100.0	100.0	100.0
road_surface	94.2	31.5	94.0	94.2	98.4	93.7	94.5	94.2
OVERALL	78.3	70.8	66.3	50.7	60.4	63.2	76.5	76.4

Yellow = withheld from training (zero-shot for that run). Per_city eval = 6984; seen_unseen eval = 8831. All entries are best/last reported epoch (PC ep6, SU ep8, all splits ep5 except Split-2a ep1).

4 Trained-task delta vs PC-base

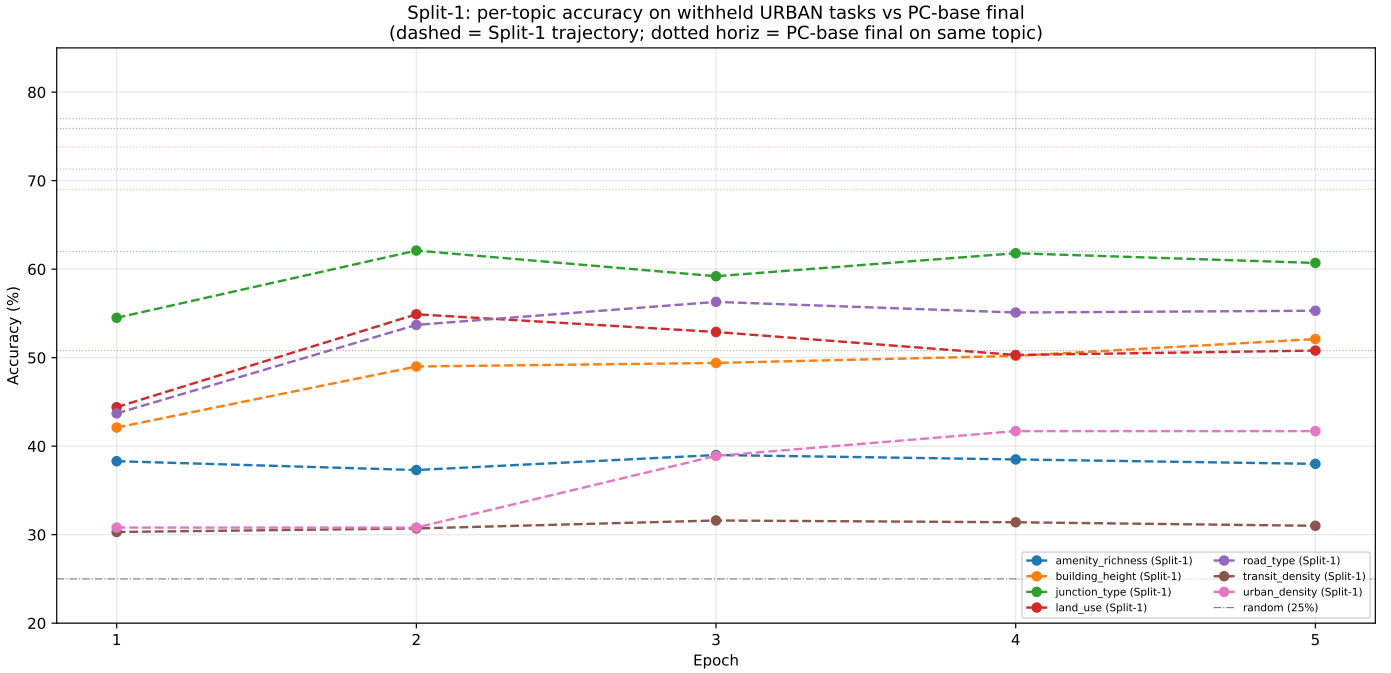
Counter-hypothesis check: does focusing the model on fewer tasks lift the kept tasks above the full-baseline?



- Joint training beats focused training on every kept-task subset.
- Smaller withholdings (Split-5,6) cost only ~2 pp on kept tasks; large ones (Split-2a) cost ~18 pp via training collapse.
- Model is not capacity-limited per-task; cross-topic gradient mixing helps.

Run	N kept	Mean Δ (pp)	Worst topic (Δ)
Split-1	7	-1.6	camera_direction (-10.5)
Split-2a	9	-17.6	junction_type (-25.0); urban_density (-19.6)
Split-3	10	-3.6	camera_direction (-13.0); land_use (+3.1)
Split-5	13	-1.8	amenity_richness (-1.0); urban_density (+2.5)
Split-6	12	-2.0	camera_direction (-12.3); land_use (+4.8)

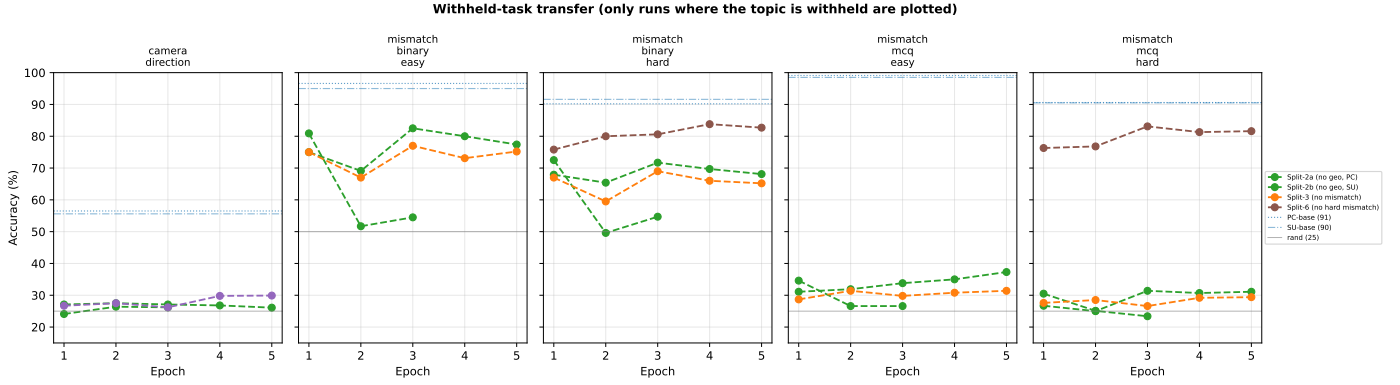
5 Withheld urban tasks (Split-1)



Topic	Split-1	PC-base	Δ	vs random
amenity_richness	38.0	62.0	-24.0	+13.0
building_height	52.1	69.0	-16.9	+27.1
junction_type	60.7	77.0	-16.3	+35.7
land_use	50.8	75.9	-25.1	+25.8
road_type	55.3	71.3	-16.0	+30.3
transit_density	31.0	50.8	-19.8	+6.0
urban_density	41.7	73.8	-32.1	+16.7
Mean	47.1	68.5	-21.4	+22.1

- Every urban topic lands between random and baseline; never above.
- transit_density: +6 over random — weakest transfer.
- urban_density: -32 pp gap — needs direct supervision.
- Most urban topics plateau by ep3-4. More training does not close the gap.

6 Withheld mismatch + camera_direction (Split-2a/2b/3/5/6)



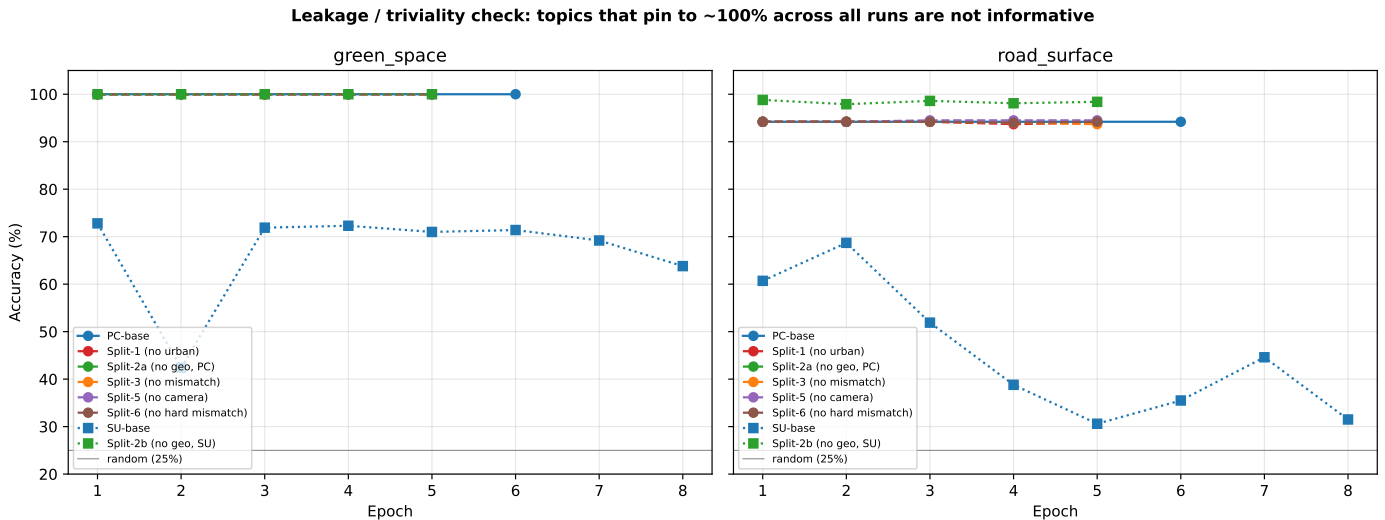
- mismatch_mcq_*: collapses to chance ($\sim 23\text{--}37\%$, random=25) when the whole family is withheld (2a/2b/3). Unlearnable from urban supervision.
- mismatch_binary_*: 65–77% under family-withholding (random=50, baseline 90+). Partial transfer.
- camera_direction (full withhold, 2a/2b/5): stalls at 26–30%, $\sim$ random. Zero transfer from any other supervision in this dataset.
- **Split-6: hard-only withholding nearly recovers:** mismatch_binary_hard 82.7% (vs 54.7% in 2a, 65.2% in 3); mismatch_mcq_hard 81.6% (vs 23.4% in 2a, 29.4% in 3). Easy variants of the same family carry the hard ones.
- Split-2b recovers more than Split-2a on mismatch (77.4 vs 54.5 binary_easy) — larger train set + no early-stop collapse.
- Split-3 final (ep5) confirms ep3 trend: mismatch_mcq_easy 31.4, hard 29.4 — camera_dir supervision does not help mismatch_mcq.

7 Saturation

Run	Last delta	Plateau ep	Trend	Auto verdict
PC-base	ep5→6: +1.4	not reached	climbing	“still improving”
SU-base	ep7→8: −1.4	ep4 (72.4)	late drift	“plateauing”
Split-1	ep4→5: +0.2	ep4	plateaued	“still improving” (misleading)
Split-2a	ep1 best (59.5)	regressed	collapsed	“still improving” (wrong)
Split-2b	ep4→5: −0.5	ep3 (61.8)	plateau	“plateauing”
Split-3	ep4→5: +0.6	ep3–5	flat	“still improving” (marginal)
Split-5	ep4→5: +0.6	ep3–5	flat	“still improving” (marginal)
Split-6	ep4→5: +0.1	ep3–5	flat	“still improving” (marginal)

- Auto verdict only checks last-epoch delta. It misses Split-2a’s collapse. Use the trajectory plots.

8 Leakage flags



- green_space: 100% in every per_city run, including ablations. SU drops to 63.8 — per_city behavior is leakage.
- road_surface: pinned at 94.2 across consecutive epochs in PC — non-vision prior. SU tanks to 31.5.
- Either drop these from headline averages or report “informative-topic average”.

	PC-base	SU-base	Split-1	Split-2a	Split-2b	Split-3	Split-5	Split-6
Mean over 12 non-trivial topics	73.7	65.6	60.9	47.0	56.4	59.6	73.0	74.1

PC-base advantage over SU-base persists (+8.1 pp). Split-5/6 sit nearly even with PC-base on non-trivial topics — single-topic withholdings have small overall cost.

9 Per-run verdicts: how trained tasks influenced unseen tasks

For each run, “trained” = topics with supervision in that run; “withheld” = topics held out and evaluated zero-shot. Δ **vs PC-base** is the gap on the same withheld topic between this run’s zero-shot performance and PC-base’s supervised performance — it isolates the transfer cost of withholding. Δ **vs random** measures whether transfer beats chance.

PC-base (full, 14 topics, 6 ep) — reference

- No withholding. 78.3% overall, still climbing at ep6.
- Sets the ceiling every ablation is measured against. Mean of 12 informative topics: 73.7%.
- amenity_richness, transit_density are the hardest *trained* topics (62, 51) — training does not solve them. urban_density jumps to 75.9 only by ep5 then drops.

SU-base (full, seen_unseen, 8 ep) — cross-city ceiling

- All 14 trained, but evaluated on held-out cities. 70.8% overall (best ep4: 72.4).
- road_surface and green_space drop hard (94.2→31.5; 100→63.8): per_city pinning is leakage; SU exposes it.
- transit_density falls 50.8→36.9 cross-city; urban_density holds at 70.6.
- Verdict: SU-base is the honest headline. PC-vs-SU gap = +7.5 pp $\approx$ pure city-leakage premium.

Split-1 (withheld 7 urban) — can geo supervision teach urban?

Trained: 7 geo (camera_dir, green_space, road_surface, 4 mismatch). Withheld: 7 urban.

- Overall 66.3 (vs PC-base 78.3). Trained tasks intact ($\geq$ baseline on all 7 geo).
- **Withheld urban: −21.4 pp on average but +22 pp over random.** Partial transfer, never recovery.
- Best transfer: junction_type (60.7, −16.3 pp), road_type (55.3, −16.0). Both lean on visual scene understanding shared with road_surface and mismatch.
- Worst transfer: transit_density (31.0, $\sim$ random) — requires explicit supervision; geo training gives *nothing*.
- urban_density: 41.7 (−32.1 pp). The single biggest transfer failure.
- amenity_richness flat at 38% all 5 ep (no learning curve at all on this withheld topic).
- **Verdict: geo supervision $\rightarrow$ urban tasks is shallow. Transfer happens for spatial-layout topics, fails for count/density topics.**

Split-2a (withheld 5 geo, per_city) — can urban supervision teach geo?

Trained: 9 urban + green_space + road_surface. Withheld: camera_dir + 4 mismatch.

- Overall 50.7 (best ep1 = 59.5, then collapse). Worst-trained run.
- Multiple *trained* topics regressed 10–25 pp: junction_type 67.9→52, urban_density 62→54, mismatch_binary_easy 80.9→54.5. mismatch_binary_easy was still in train set, so this is genuine training instability, not transfer.

- Withheld camera_dir: 26.2 (random=25). Zero transfer.
- Withheld mismatch_mcq*: 23–27% (random=25). Zero transfer.
- Withheld mismatch_binary*: 54.5–54.7% (random=50). $\sim +5$ pp over random — minimal.
- **Verdict: removing both mismatch and camera together destabilizes training; transfer to either is essentially nil. Do not interpret as “urban knowledge does not help geo” — the run never converged.**

Split-2b (withheld 5 geo, seen_unseen) — same withholding, larger / more diverse train set

- Overall 60.4 (best ep3 = 61.8). Stable, unlike 2a.
- Withheld camera_dir: 26.1 (random=25). Same zero transfer as 2a.
- Withheld mismatch_binary: 68–77% (vs 2a 55%, baseline 90+). Better than 2a because training did not collapse, but still ~ 20 pp below baseline.
- Withheld mismatch_mcq: 31–37% (random=25). Slow climb over epochs — some weak transfer; still far from baseline 90%.
- **Verdict: with stable training, urban supervision lifts mismatch_binary modestly and mismatch_mcq slightly above chance. camera_direction is unreachable from urban supervision regardless.**

Split-3 (withheld 4 mismatch, camera_dir kept) — isolates the mismatch family

Trained: 9 urban + green_space + road_surface + camera_direction. Withheld: 4 mismatch*.

- Overall 63.2 (5 ep). Trained tasks ~ -3 pp vs PC-base on average; camera_dir lands at 43.5 vs PC-base 56.5 (-13 pp) — camera_dir benefits from mismatch supervision being present.
- Withheld mismatch_binary: 65.2–75.2% (random=50). Partial transfer.
- Withheld mismatch_mcq: 29–31% (random=25). **Stuck at chance for all 5 epochs** — camera_dir supervision does not unlock mismatch_mcq.
- Cross-check vs Split-2a: mismatch_binary in Split-3 (65–75) > Split-2a (55) but $\approx$ Split-2b (68–77). Split-2b’s improvement was training stability, not camera_dir.
- **Verdict: mismatch_mcq is independently structured; no transfer source in this dataset reaches it. mismatch_binary gets ~ 15 – 25 pp from any non-collapsed training, regardless of which other tasks are present.**

Split-5 (withheld camera_direction only) — isolates camera_dir transfer

Trained: 13 (everything except camera_direction). Withheld: camera_direction.

- Overall 76.5 (vs PC-base 78.3). Smallest training-set delta of any ablation.
- Trained tasks ~ -1.8 pp vs PC-base on average — closest to baseline of any ablation.
- Withheld camera_direction: 26.7 $\rightarrow$ 29.9 over 5 epochs (random=25, baseline 56.5). **$\sim$ random, -26.6 pp vs baseline.**
- This is the cleanest measurement we have: with 13 supervised topics including all 4 mismatch + green_space + road_surface + 7 urban, camera_direction still doesn’t transfer at all.

- **Verdict:** camera_direction has zero transfer source in this dataset. It must be supervised directly. The other 13 topics share no useful representation for it.

Split-6 (withheld hard mismatch only) — does the easy variant teach the hard one?

Trained: 12 (drops mismatch_binary_hard + mismatch_mcq_hard). Withheld: 2 hard mismatch.

- Overall 76.4 (vs PC-base 78.3). Tied with Split-5 for smallest cost.
- Withheld mismatch_binary_hard: **82.7%** (random=50, baseline 90.2, −7.5 pp). Strong transfer.
- Withheld mismatch_mcq_hard: **81.6%** (random=25, baseline 90.6, −9.0 pp). Strong transfer.
- Compare to Split-2a (whole family withheld): hard_binary 54.7, hard_mcq 23.4. Easy-variant supervision recovers ~28–58 pp on the hard variant.
- **Verdict:** easy and hard mismatch share representation almost completely. Hard variants are not a separate learnable skill — they are easy variants under harder distractors. This is the strongest positive transfer result in the study.

Cross-cutting transfer summary

Withheld topic	Run	Withheld acc	vs PC-base	vs random
camera_direction	Split-5 (only that)	29.9	−26.6	+4.9
camera_direction	Split-2a (with mismatch)	26.2	−30.3	+1.2
camera_direction	Split-2b (SU)	26.1	−29.5	+1.1
mismatch_mcq_easy	Split-3 (whole family out)	31.4	−67.7	+6.4
mismatch_mcq_easy	Split-2a	26.6	−72.5	+1.6
mismatch_mcq_easy	Split-2b (SU)	37.3	−61.2	+12.3
mismatch_mcq_hard	Split-3	29.4	−61.2	+4.4
mismatch_mcq_hard	Split-2a	23.4	−67.2	−1.6
mismatch_mcq_hard	Split-2b (SU)	31.1	−59.4	+6.1
mismatch_mcq_hard	Split-6 (only hard out)	81.6	−9.0	+56.6
mismatch_binary_easy	Split-3	75.2	−21.4	+25.2
mismatch_binary_easy	Split-2a	54.5	−42.1	+4.5
mismatch_binary_easy	Split-2b (SU)	77.4	−17.6	+27.4
mismatch_binary_hard	Split-3	65.2	−25.0	+15.2
mismatch_binary_hard	Split-2a	54.7	−35.5	+4.7
mismatch_binary_hard	Split-6 (only hard out)	82.7	−7.5	+32.7
urban_density	Split-1	41.7	−32.1	+16.7
land_use	Split-1	50.8	−25.1	+25.8
amenity_richness	Split-1	38.0	−24.0	+13.0
junction_type	Split-1	60.7	−16.3	+35.7
road_type	Split-1	55.3	−16.0	+30.3
transit_density	Split-1	31.0	−19.8	+6.0
building_height	Split-1	52.1	−16.9	+27.1

- **Three transfer regimes are visible:**
 - *Strong (within-family):* easy→hard mismatch (Split-6): −7 to −9 pp gap, +30 to +57 over random.
 - *Partial (cross-family, structured):* urban tasks under geo supervision (Split-1, Split-2b): −16 to −25 pp, +10 to +35 over random.

- *None*: camera_direction under any supervision; mismatch_mcq when whole family withheld — both essentially at random.
- Topics with *strong cross-family* transfer in this dataset: junction_type, road_type (mostly visual scene cues).
- Topics with *no transfer source*: camera_direction, transit_density, mismatch_mcq.
- Practical implication: when curating a smaller training mix, the *family* matters more than the count. Dropping one topic from a family is cheap; dropping the whole family is catastrophic.

10 For the paper

Report:

- Joint training beats focused training on every kept-task subset.
- Three transfer regimes: strong within-family (easy→hard mismatch, -8 pp), partial cross-family for spatial-layout tasks (-16 to -25 pp), none for camera_direction or mismatch_mcq when family withheld ($\sim$ random).
- mismatch_mcq + camera_direction need direct supervision; they are not transfer targets.
- Urban tasks transfer partially: $+22$ pp over random when withheld, but ~ 20 pp below baseline.
- Cross-city eval costs $\sim 7-8$ pp uniformly. Use SU-base (70.8%) as the honest headline.
- green_space, road_surface are leaked on per_city. Flag or exclude.

Do NOT claim:

- “Urban knowledge transfers to mismatch tasks” — mismatch_mcq stays at chance whenever the whole family is withheld.
- “camera_direction transfers” — Split-5 isolates this and shows $\sim$ random with all 13 other topics supervised.
- “Ablation X is close to baseline” — different denominator topic sets.
- Cite the auto verdict banner — it is wrong for Split-2a.

Suggested compact paper table:

Train mix	Eval	Overall	Trained mean	Withheld mean	vs random
all 14 (PC-base)	per_city	78.3	78.3	—	—
all 14 (SU-base)	seen_unseen	70.8	70.8	—	—
geo only (Split-1)	per_city	66.3	91.4	47.1	+22
urban only PC (Split-2a)	per_city	50.7	65.4	37.0	$\sim +0$
urban only SU (Split-2b)	seen_unseen	60.4	70.4	48.0	$\sim +10$
urban+cam (Split-3)	per_city	63.2	76.0	50.4	+8
no camera (Split-5)	per_city	76.5	80.1	29.9	+5
no hard mismatch (Split-6)	per_city	76.4	79.6	82.2	+32

mismatch_binary random=50; mismatch_mcq, camera=25. Withheld mean uses only the topics dropped from that run.

A Per-epoch trajectories

PC-base (per_city, 6 ep)

	ep1	ep2	ep3	ep4	ep5	ep6
Overall	71.5	72.3	75.0	76.4	76.9	78.3
amenity_richness	49.2	49.2	51.9	57.2	53.5	62.0
building_height	62.5	60.2	60.2	65.1	64.4	69.0
camera_direction	25.3	35.1	46.0	51.0	53.1	56.5
green_space	100.0	100.0	100.0	100.0	100.0	100.0
junction_type	69.9	71.7	72.8	76.6	70.1	77.0
land_use	76.8	75.9	76.5	77.0	77.5	75.9
mismatch_binary_easy	91.6	93.6	96.3	94.5	96.8	96.6
mismatch_binary_hard	87.2	85.0	88.6	88.8	92.0	90.2
mismatch_mcq_easy	95.0	94.3	96.8	98.2	98.0	99.1
mismatch_mcq_hard	82.7	81.5	86.8	88.2	90.6	90.6
road_surface	94.2	94.2	94.2	94.2	94.2	94.2
road_type	73.3	71.1	73.1	74.7	71.3	71.3
transit_density	44.6	44.9	44.6	47.2	48.5	50.8
urban_density	65.1	69.5	72.5	68.3	75.9	73.8

SU-base (seen_unseen, 8 ep, early-stopped)

	ep1	ep2	ep3	ep4	ep5	ep6	ep7	ep8
Overall	67.2	66.0	70.7	72.4	71.9	72.5	72.2	70.8
amenity_richness	57.8	35.7	61.6	65.4	54.5	63.6	61.6	59.4
building_height	44.1	43.8	51.8	55.9	54.0	54.8	52.6	54.4
camera_direction	26.5	28.7	42.5	48.0	55.5	57.0	56.7	55.6
green_space	72.8	42.4	71.9	72.3	71.0	71.4	69.2	63.8
junction_type	72.7	68.3	67.5	68.1	75.6	69.4	69.2	71.4
land_use	70.8	72.9	69.1	74.2	74.7	71.4	71.0	69.1
mismatch_binary_easy	93.5	93.5	94.0	93.8	96.2	95.9	97.3	95.0
mismatch_binary_hard	81.5	81.2	86.3	89.7	89.0	91.1	92.6	91.6
mismatch_mcq_easy	89.6	95.8	95.3	97.6	98.1	98.0	98.1	98.5
mismatch_mcq_hard	78.9	82.9	81.9	87.4	87.7	88.9	91.1	90.5
road_surface	60.7	68.7	51.9	38.8	30.6	35.5	44.6	31.5
road_type	71.4	72.3	70.0	70.8	73.1	70.4	67.7	70.9
transit_density	46.5	38.7	49.5	50.6	47.2	46.5	41.3	36.9
urban_density	62.5	70.1	76.5	75.6	70.9	72.4	70.4	70.6

road_surface degrades 60.7→31.5 over 8 epochs — model overfits per-city prior, breaks cross-city.

Split-1 (no urban, 5 ep)

	ep1	ep2	ep3	ep4	ep5
Overall	59.6	63.4	64.8	66.1	66.3
amenity_richness	38.3	37.3	39.0	38.5	38.0
building_height	42.1	49.0	49.4	50.2	52.1
camera_direction	26.0	31.4	40.5	41.9	46.0
green_space	100.0	100.0	100.0	100.0	100.0
junction_type	54.5	62.1	59.2	61.8	60.7
land_use	44.4	54.9	52.9	50.3	50.8
mismatch_binary_easy	94.5	96.4	92.7	98.0	97.1
mismatch_binary_hard	84.0	89.1	87.5	92.7	92.3
mismatch_mcq_easy	93.6	95.2	97.1	97.9	98.0
mismatch_mcq_hard	79.3	83.1	86.1	88.4	88.2
road_surface	94.2	94.2	94.2	93.7	94.0
road_type	43.7	53.7	56.3	55.1	55.3
transit_density	30.3	30.7	31.6	31.4	31.0
urban_density	30.8	30.8	38.9	41.7	41.7

transit_density flat 30–31% all 5 epochs; amenity_richness flat ~38%. Geo-only training has zero signal here.

Split-2a (no geo, PC — regressed)

	ep1	ep2	ep3
Overall	59.5	50.4	50.7
amenity_richness	50.4	36.2	36.9
building_height	60.5	44.4	43.7
camera_direction	24.1	26.4	26.2
green_space	100.0	100.0	100.0
junction_type	67.9	58.3	52.0
land_use	76.3	72.9	73.6
mismatch_binary_easy	80.9	51.7	54.5
mismatch_binary_hard	72.5	49.6	54.7
mismatch_mcq_easy	34.6	26.6	26.6
mismatch_mcq_hard	30.5	25.1	23.4
road_surface	94.2	94.2	94.2
road_type	72.0	66.8	67.6
transit_density	39.9	34.9	36.7
urban_density	62.2	55.1	54.2

ep1 was best; multiple trained tasks regressed 10+ pp afterward.

Split-2b (no geo, SU)

	ep1	ep2	ep3	ep4	ep5
Overall	58.9	59.0	61.8	60.9	60.4
amenity_richness	58.1	61.7	62.9	60.4	59.7
building_height	47.1	50.4	51.1	53.7	54.0
camera_direction	27.1	27.5	27.1	26.8	26.1
green_space	100.0	100.0	100.0	100.0	100.0
junction_type	75.2	74.7	72.9	69.4	69.6
land_use	70.4	74.6	74.0	75.6	76.2
mismatch_binary_easy	75.0	69.1	82.5	80.0	77.4
mismatch_binary_hard	67.9	65.4	71.7	69.7	68.1
mismatch_mcq_easy	31.1	31.9	33.8	35.0	37.3
mismatch_mcq_hard	26.7	25.0	31.4	30.7	31.1
road_surface	98.8	97.9	98.6	98.1	98.4
road_type	73.1	71.7	72.7	71.3	72.1
transit_density	45.7	44.9	49.1	48.8	44.9
urban_density	71.3	75.0	75.8	73.3	72.9

Best ep3 (61.8); mild -1.4 drift. mismatch_mcq slowly climbs ep1 31.1 $\rightarrow$ ep5 37.3 on transfer alone.

Split-3 (no mismatch, 5 ep)

	ep1	ep2	ep3	ep4	ep5
Overall	57.8	59.3	62.7	62.6	63.2
amenity_richness	52.6	53.7	61.3	62.2	62.6
building_height	49.8	62.8	64.8	66.3	66.3
camera_direction	23.0	33.0	42.8	39.9	43.5
green_space	100.0	100.0	100.0	100.0	100.0
junction_type	66.1	71.4	74.6	75.4	76.6
land_use	76.1	76.5	77.9	80.7	79.0
mismatch_binary_easy	75.0	67.0	77.0	73.1	75.2
mismatch_binary_hard	67.0	59.5	69.0	66.0	65.2
mismatch_mcq_easy	28.7	31.4	29.8	30.8	31.4
mismatch_mcq_hard	27.6	28.5	26.6	29.2	29.4
road_surface	94.2	94.2	94.2	94.0	93.7
road_type	70.9	73.6	72.7	72.7	72.7
transit_density	45.6	44.6	47.4	46.2	49.0
urban_density	63.1	69.3	72.5	72.9	73.3

mismatch_mcq stays near chance ($\sim 29\%$) for all 5 epochs even with camera_direction supervised. mismatch_binary stuck at $\sim 65\text{--}75\%$.

Split-5 (no camera_direction, 5 ep)

	ep1	ep2	ep3	ep4	ep5
Overall	70.7	71.8	74.9	75.9	76.5
amenity_richness	51.7	50.6	55.6	59.0	61.0
building_height	54.4	61.3	61.7	65.5	65.1
camera_direction	26.7	27.5	26.2	29.8	29.9
green_space	100.0	100.0	100.0	100.0	100.0
junction_type	69.0	72.3	75.9	77.0	78.3
land_use	75.6	75.6	78.6	77.5	77.9
mismatch_binary_easy	92.0	96.1	96.1	97.1	97.3
mismatch_binary_hard	86.5	83.6	90.0	91.4	92.2
mismatch_mcq_easy	95.7	94.3	98.4	98.2	98.4
mismatch_mcq_hard	81.6	82.7	88.4	90.6	90.2
road_surface	94.2	94.2	94.5	94.5	94.5
road_type	70.4	70.6	73.6	73.4	74.0
transit_density	44.6	45.5	48.3	48.0	48.1
urban_density	61.1	66.5	72.9	73.8	76.3

camera_direction trajectory: 26.7→29.9 over 5 epochs (random=25, baseline 56.5). Overall climbs +5.8 pp on the other 13 topics — camera_dir is not pulled along by capacity gains.

Split-6 (no hard mismatch, 5 ep)

	ep1	ep2	ep3	ep4	ep5
Overall	68.5	73.3	75.1	76.3	76.4
amenity_richness	47.8	58.6	58.3	59.9	61.9
building_height	61.3	57.9	62.5	65.9	68.2
camera_direction	25.8	41.7	39.9	45.1	44.2
green_space	100.0	100.0	100.0	100.0	100.0
junction_type	65.8	72.1	74.6	76.3	75.9
land_use	77.4	76.3	79.9	79.0	80.7
mismatch_binary_easy	93.2	95.7	96.8	97.3	97.3
mismatch_binary_hard	75.8	80.0	80.6	83.8	82.7
mismatch_mcq_easy	93.0	95.5	97.0	98.0	97.5
mismatch_mcq_hard	76.3	76.8	83.1	81.3	81.6
road_surface	94.2	94.2	94.2	94.0	94.2
road_type	66.7	70.1	73.3	73.6	73.6
transit_density	39.8	47.6	50.4	51.7	51.0
urban_density	61.0	71.7	72.2	74.0	74.3

Withheld hard variants reach 81–83% by ep5 (vs $\sim 90\%$ in PC-base) — only ~ 8 pp gap. The easy variant of the same family carries the hard one almost fully.

B Method

- Ablation runs: bs=96, NUM_EPOCHS=5, early-stop patience=2, cosine LR, 100 warmup, LoRA $r=16/\alpha=16$, all-linear, SEED=3407, RTX Pro 6000 96 GB.
- PC-base bs=86, 6 ep; SU-base bs=80, 8 ep — earlier runs, same pipeline, identical eval routine.
- Eval: greedy decoding, deterministic. PC=6984 samples, SU=8831.
- Baselines per topic from verdict files: mismatch_binary 2-way (rand=50), all others 4-way (rand=25).